

FINAL REPORT

Global Access to Clean Fuels and Technologies for Cooking (2016–2020): A Power BI Dashboard Analysis

1. Objective / Problem Statement

Access to clean fuels and technologies for cooking is essential for improving public health, reducing environmental pollution, and achieving the United Nations Sustainable Development Goal (SDG) 7: Affordable and Clean Energy. Despite global efforts, access to clean cooking fuels varies significantly across countries.

The objective of this project is to analyze global access to clean fuels and technologies for cooking between 2016 and 2020, identify trends and disparities among countries, and present meaningful insights through an interactive Power BI dashboard.

2. Dataset Description

The dataset used for this project was obtained from the World Development Indicators (WDI) database published by the World Bank.

Dataset Details

- **Source:** World Development Indicators (World Bank)
- **Indicator:** Access to Clean Fuels and Technologies for Cooking (% of Population)
- **Period Covered:** 2016–2020
- **Countries/Territories:** 265
- **Years:** 5

Variables Used

- Country Name
- Country Code
- Year
- Indicator Value (Access Rate)

The dataset measures the percentage of the population in each country with access to clean fuels and technologies for cooking.

3. Data Cleaning Process

The dataset was cleaned and transformed using SQL and Power Query before building the Power BI dashboard. The following steps were performed:

- 1 Used SQL to extract only the “**Access to Clean Fuels and Technologies for Cooking**” indicator for the years 2016–2020.
- 2 Filtered the dataset to retain only the required records for analysis.
- 3 Imported the extracted dataset into Power BI.
- 4 Used Power Query to unpivot the year columns (2016, 2017, 2018, 2019, and 2020) to convert the dataset from a wide format into a long format.
- 5 Renamed the generated columns: Attribute → Year, Value → Indicator Value.
- 6 Converted the Year column to Whole Number and the Indicator Value column to Decimal Number.
- 7 Checked for and removed duplicate records.
- 8 Checked for and handled missing or null values where necessary.
- 9 Standardized country names and country codes to ensure consistency.
- 10 Created an Access Category column using DAX to classify countries into: 0–25%, 26–50%, 51–75%, and 76–100%.
- 11 Loaded the cleaned dataset into the Power BI data model for visualization and analysis.

4. Analysis Methodology

The analysis was carried out using Power BI. The following techniques were applied:

- Data transformation using Power Query
- DAX calculations for creating KPIs and categorizing access levels
- KPI cards to summarize key performance metrics
- Line chart to analyze yearly trends
- Donut chart to display the distribution of countries by access category
- Bar charts to compare countries with the highest and lowest access rates
- Matrix table to compare yearly access rates across countries
- Slicers for Year and Country Code to enable interactive dashboard filtering

The following KPIs were calculated:

- Total Number of Countries
- Average Access Rate
- Highest Access Rate
- Lowest Access Rate
- Number of Years Covered

5. Key Findings

The dashboard analysis revealed the following:

- A total of 265 countries and territories were analyzed.
- The average global access rate was 67.05%.
- The highest access rate recorded was 100.00%.
- The lowest access rate recorded was 0.10%.
- The analysis covered five years (2016–2020).
- Global access to clean cooking fuels and technologies increased steadily throughout the study period.
- Most countries fall within the 76–100% access category.
- Countries such as Andorra, Australia, Austria, Belgium, and Canada consistently achieved universal or near-universal access.
- Countries including Guinea, Central African Republic, Sierra Leone, Liberia, and Burundi recorded the lowest average access rates.

6. Insights

The analysis provides the following insights:

- Global access to clean cooking fuels and technologies improved consistently from 2016 to 2020, indicating positive progress toward universal clean energy access.
- Although many countries have achieved universal access, significant disparities still exist between developed and developing countries.
- Most countries fall within the 76–100% access category, showing that a large proportion of countries have relatively high access to clean cooking technologies.

- Countries with extremely low access rates require greater investment in clean energy infrastructure and supportive government policies.
- The large gap between countries with 100% access and those with access below 1% highlights the unequal distribution of clean cooking technologies worldwide.

7. Recommendations

Based on the findings, the following recommendations are proposed:

- 1 Increase investment in clean cooking infrastructure, especially in countries with low access rates.
- 2 Expand government subsidies and incentives to make clean cooking technologies more affordable.
- 3 Strengthen partnerships between governments, international organizations, and private-sector stakeholders to improve access.
- 4 Promote public awareness of the health, environmental, and economic benefits of clean cooking fuels.
- 5 Continue monitoring progress using data analytics and interactive dashboards to support evidence-based policy decisions.
- 6 Prioritize interventions in countries with the lowest access rates to accelerate progress toward achieving Sustainable Development Goal (SDG) 7 – Affordable and Clean Energy.

8. Conclusion

This project analyzed global access to clean fuels and technologies for cooking from 2016 to 2020 using data from the World Development Indicators (WDI) and presented the findings through an interactive Power BI dashboard. The analysis showed that global access improved steadily during the study period, with an average access rate of 67.46% across 265 countries and territories. While many countries achieved universal access, significant disparities remain, particularly in low-income countries where access is still extremely limited.

The dashboard effectively communicates these trends through KPIs, charts, and interactive filters, enabling users to explore country-level performance and global patterns. Overall, the findings emphasize the need for continued investment, supportive government policies, and international collaboration to ensure universal access to clean cooking fuels and technologies, thereby improving public health, protecting the environment, and advancing the achievement of Sustainable Development Goal 7 (Affordable and Clean Energy).